

Practical 5

About this unit

Practical 5

Practice Lab Assignment

Unit • 100% completed



Lab Assignment

Unit • 100% completed



Course

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5.1.1. Stacked Plot21.33

Create a stacked area plot to visualize the temperature variations for three different cities (City A, City B, and City C) across the months of the year. The temperature data is provided for each city in the editor.

Your task is to:

- Create a stacked area plot using the data.
- Label the x-axis as "Month", the y-axis as "Temperature", and provide the title "Temperature Variation" for the plot.
- Display the plot showing the temperature variation for each city throughout the months of the year.

Sample Test Cases

stackedpl...

```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 # Data for Months and Temperature for three cities
5 data = {
6     'Month': ['January', 'February', 'March', 'April', 'May', 'June',
7             'July', 'August', 'September', 'October', 'November', 'December'],
8     'City_A_Temperature': [5, 7, 10, 13, 17, 20, 22, 21, 18, 12, 8, 6],
9     'City_B_Temperature': [2, 3, 5, 6, 10, 14, 16, 17, 12, 9, 5, 3],
10    'City_C_Temperature': [3, 4, 6, 8, 9, 12, 15, 14, 10, 7, 4, 2]
11 }
12
13 # Write your code...
14 df = pd.DataFrame(data)
15 plt.stackplot(df["Month"], df["City_A_Temperature"], df["City_B_Temperature"], df["City_C_Temperature"])
16 plt.title("Temperature Variation")
17 plt.xlabel("Month")
18 plt.ylabel("Temperature")
19 plt.show()
```

Average time0.280 s280.00 ms

Maximum time0.280 s280.00 ms

1 out of 1 shown test case(s) passed

Terminal

Test cases

27°C Sunny

Search

09:12 07-05-2026

Course

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5.2.1. Titanic Dataset

02:06

Write a Python program to analyze and visualize data from the Titanic dataset based on the following instructions:

Dataset Information:

The dataset is stored in a CSV file named `titanic.csv` and has been loaded using the `pandas` library. It contains the following columns:

- `Pclass`: Passenger class (1 = First, 2 = Second, 3 = Third).
- `Gender`: Gender of the passenger (male/female).
- `Age`: Age of the passenger.
- `Survived`: Survival status (0 = Did not survive, 1 = Survived).
- `Fare`: Ticket fare paid by the passenger.

Visualization:

To represent these trends, you will create 5 visualizations using Matplotlib. The visualizations should be arranged in a 3x2 grid (3 rows and 2 columns).

Visualization Details:

Write the code to create a series of visualizations as follows:

Bar Plot (Pclass Distribution):

- Create a bar plot to show the distribution of passengers across the different passenger classes (`Pclass`).
- Use the color `skyblue` for the bars.

titanicDat...

Submit

6

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17

18

19

Set up the figure for 5 subplots

fig, axes = plt.subplots(3, 2, figsize=(12, 12))

write the code..

count=df['Pclass'].value_counts().sort_index()

count_gen=df['Gender'].value_counts()

count_s=df['Survived'].value_counts().sort_index()

axes[0,0].bar(count.index,count.values,color='skyblue'))

axes[0,0].set_title('Passenger Class Distribution')

axes[0,0].set_xlabel('Pclass')

axes[0,0].set_ylabel('Count')

Average time

1.210 s

1210.00 ms

Maximum time

1.210 s

1210.00 ms

1 out of 1 shown test case(s) passed

Test case 1

1210 ms

Debug

Expected output

Actual output

Terminal

Test cases

27°C Sunny

Search

09:13 07-05-2026

5.2.2. Histogram of passenger information of Titanic

10/12

Write a Python code to plot a histogram for the distribution of the 'Age' column from the Titanic dataset. The histogram should display the frequency of different age ranges with the following specifications:

1. Use **30 bins** for the histogram.
2. Set the **edge color** of the bars to **black (k)**.
3. Label the x-axis as **'Age'** and the y-axis as **'Frequency'**.
4. Add the title **"Age Distribution"** to the histogram.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,0,5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,3 340900,21.075,C
```

Histogram...

Submit

```
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Histogram
17 plt.hist(data["Age"], bins=30, edgecolor='k')
18 plt.xlabel("Age")
19 plt.ylabel("Frequency")
20 plt.title("Age Distribution")
21 plt.show()
```

Average time
0.369 s
369.00 ms

Maximum time
0.369 s
369.00 ms

1 out of 1 shown test case(s) passed

Test case 1 369 ms Debug

Expected output

Actual output

Terminal

Test cases

5.2.3. Bar plot of survival rate of passengers

Write a Python code to plot a bar chart that shows the count of passengers who survived and did not survive in the Titanic dataset. The chart should display the following specifications:

1. Use the 'Survived' column to show the count of survivors (0 = Did not survive, 1 = Survived).

2. Set the chart type to 'bar'.

3. Add the title 'Survival Count' to the chart.

4. Label the x-axis as 'Survived' and the y-axis as 'Count'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Morian, Mr. James",male,,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,2,3,349909,21.075,C

BarPlotOf...

7

Data Cleaning

8

data['Age'].fillna(data['Age'].median(), inplace=True)

9

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10

data.drop('Cabin', axis=1, inplace=True)

11

12

Convert categorical features to numeric

13

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16

Write your code here for Bar Plot for Survival Rate

17

sc = data['Survived'].value_counts()

18

sc.plot(kind='bar')

19

20

plt.xlabel("Survived")

21

plt.ylabel("Count")

22

plt.title("Survival Count")

23

24

plt.show()

Average time

0.453 s

453.00 ms

Maximum time

0.453 s

453.00 ms

1 out of 1 shown test case(s) passed

Test case 1

453 ms

Debug

Expected output

Actual output

Terminal

Test cases

5.2.4. Bar Plot for Survival by Gender

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by gender, in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the 'Sex' column, then use the value_counts() function to count the occurrences of survivors (0 = Did not survive, 1 = Survived) for each gender.

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Gender" to the chart.

4. Label the x-axis as 'Gender' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

BarPlotOf...

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Bar Plot for Survival by Gender

17 sc = data.groupby('Sex')['Survived'].value_counts().unstack()

18 sc.plot(kind='bar', stacked=True)

19

20

21 plt.xlabel("Gender")

22 plt.ylabel("Count")

23 plt.title("Survival by Gender")

24 plt.legend(['Not Survived', 'Survived'])

25 plt.show()

Average time 0.507 s 607.00 ms

Maximum time 0.507 s 607.00 ms

1 out of 1 shown test case(s) passed

Test case 1 507 ms

Debug

Expected output

Actual output

Terminal

Test cases

5.2.5. Bar Plot for Survival by Pclass

Write a Python code to plot a stacked bar chart that shows the count of passengers who survived and did not survive, grouped by passenger class (Pclass), in the Titanic dataset. The chart should display the following specifications:

1. Group the data by the Pclass column and count the number of survivors (0 = Did not survive, 1 = Survived) for each class using value_counts().

2. Use a stacked bar chart to display the survival counts.

3. Add the title "Survival by Pclass" to the chart.

4. Label the x-axis as 'Pclass' and the y-axis as 'Count'.

5. The legend should indicate 'Not Survived' and 'Survived'.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22	1	0	A/5 21171	7.25	S	
2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282	7.925	S	
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803	53.1	C123	S

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

BarPlotOf...

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Bar Plot for Survival by Pclass

17

18 sc = data.groupby('Pclass')['Survived'].value_counts().unstack()

19 sc.plot(kind='bar', stacked=True)

20

21 plt.xlabel("Pclass")

22 plt.ylabel("Count")

23 plt.title("Survival by Pclass")

24 plt.legend(['Not Survived', 'Survived'])

25 plt.show()

Average time 0.478 s 478.00 ms

Maximum time 0.478 s 478.00 ms

1 out of 1 shown test case(s) passed

Test case 1 478 ms

Debug

Expected output

Actual output

Terminal

Test cases

5.2.6. Bar Plot for Survival by Embarked

Write a Python code to plot a stacked bar chart showing the survival count for passengers based on their embarkation location in the Titanic dataset.
The chart should display the following specifications:
1. Use the **Embarked** column to determine the embarkation location. After converting this column into dummy variables (using **pd.get_dummies()**), plot the survival count based on the **Embarked_Q** column (representing passengers who embarked from Queenstown) in relation to survival.
2. Set the chart type to 'bar' and make it stacked.
3. Add the title "**Survival by Embarked**" to the chart.
4. Label the x-axis as '**Embarked**' and the y-axis as '**Count**'.
5. Include a legend to distinguish between survivors and non-survivors (label the legend as '**Survived**' and '**Not Survived**').

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

BarPlotOf...

8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Bar Plot for Survival by Embarked
17
18 survival_counts=data.groupby('Embarked_Q')
19 ['Survived'].value_counts().unstack()
20 survival_counts.plot(kind='bar',stacked=True)
21 plt.title('Survival by Embarked')
22 plt.xlabel('Embarked')
23 plt.ylabel('Count')
24 plt.legend(['Not Survived','Survived'])
25 plt.show()

Average time 0.529 s
Maximum time 0.529 s
1 out of 1 shown test case(s) passed

Test case 1 0.29 ms

Terminal Test cases

5.2.7. Box plot for Age Distribution

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset across different passenger classes. The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.

2. Set the title of the plot to **"Age by Pclass"**.

3. Remove the default subtitle with **plt.suptitle('')**.

4. Label the x-axis as **'Pclass'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Dooley, Master. George Leonard",male,2,2,1,240000,71.075,,S

BoxPlotF...

7

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

11

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

Write your code here for Box Plot for Age by Pclass

data.boxplot(column = 'Age',by='Pclass')

plt.suptitle('')

19

plt.title('Age by Pclass')

plt.xlabel('Pclass')

plt.ylabel('Age')

plt.show()

23

Average time

0.481 s

481.00 ms

Maximum time

0.481 s

481.00 ms

1 out of 1 shown test case(s) passed

Test case 1

481 ms

Debug

Expected output

Actual output

Terminal

Test cases

< Prev

Reset

Submit

Next >

5.2.8. Box Plot for Age by Survived

Write a Python code to plot a boxplot that shows the distribution of the 'Age' column from the Titanic dataset based on whether passengers survived or not. The boxplot should display the following specifications:

1. Use the **Survived** column to group the data for the boxplot (0 = Did not survive, 1 = Survived).

2. Set the title of the plot to **"Age by Survival"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Survived'** and the y-axis as **'Age'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q

BoxPlotF...

7

Data Cleaning

8 data['Age'].fillna(data['Age'].median(), inplace=True)

9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

10 data.drop('Cabin', axis=1, inplace=True)

11

12 # Convert categorical features to numeric

13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

15

16 # Write your code here for Box Plot for Age by Survived

17

18 data.boxplot(column='Age', by='Survived')

19 plt.title("Age by Survival")

20 plt.suptitle('')

21

22

23 plt.xlabel('Survived')

24 plt.ylabel('Age')

25 plt.show()

Average time

0.465 s

465.00 ms

Maximum time

0.465 s

465.00 ms

1 out of 1 shown test case(s) passed

Test case 1 465 ms Debug

Terminal Test cases

< Prev Reset Submit Next >

5.2.9. Box Plot for Fare by Pclass

Write a Python code to plot a boxplot that shows the distribution of the 'Fare' column from the Titanic dataset based on the passenger class (Pclass). The boxplot should display the following specifications:

1. Use the **Pclass** column to group the data for the boxplot.

2. Set the title of the plot to **"Fare by Pclass"**.

3. Remove the default subtitle with **plt.suptitle("")**.

4. Label the x-axis as **'Pclass'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked

1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S

2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C

3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S

4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S

5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S

6,0,3,"Moran, Mr. James",male,0,0,330877,8.4583,,Q

7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S

8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S

BoxPlotF...

data = pd.read_csv('Titanic-dataset.csv')

Data Cleaning

data['Age'].fillna(data['Age'].median(), inplace=True)

data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)

data.drop('Cabin', axis=1, inplace=True)

Convert categorical features to numeric

data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})

data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

Write your code here for Box Plot for Fare by Pclass

data.boxplot(column='Fare', by='Pclass')

plt.title("Fare by Pclass")

plt.suptitle("")

plt.xlabel("Pclass")

plt.ylabel("Fare")

plt.show()

Average time 0.479 s

Maximum time 0.479 s

1 out of 1 shown test case(s) passed

Test case 1 479 ms

Terminal Test cases

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Course

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CODETANTRA

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202501090065@mitaoe.ac.in Support Logout

5.2.10. Scatter Plot for Age vs. Fare

08:57

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset. The scatter plot should display the following specifications:

- Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- Set the title of the plot to **"Age vs. Fare"**.
- Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,4/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17509,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
```

AgeFareS...

Submit

```
5 data = pd.read_csv('Titanic-Dataset.csv')
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Box Plot for Fare by Pclass
17
18 plt.scatter(data['Age'], data['Fare'])
19 plt.title('Age vs. Fare')
20 plt.xlabel('Age')
21 plt.ylabel('Fare')
22 plt.show()
```

Average time

0.329 s

Maximum time

0.329 s

329.00 ms

329.00 ms

1 out of 1 shown test case(s) passed

Test case 1

329 ms

Debug

Expected output

Actual output

Terminal

Test cases

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5.2.11. Scatter Plot for Age vs. Fare by Survived

09:02

Write a Python code to plot a scatter plot showing the relationship between the 'Age' and 'Fare' columns in the Titanic dataset, with points color-coded by survival status. The scatter plot should display the following specifications:

- Use the **Age** column for the x-axis and the **Fare** column for the y-axis.
- Color the points based on the **Survived** column: **Red** for passengers who did not survive (**Survived = 0**). **Blue** for passengers who survived (**Survived = 1**).
- Set the title of the plot to **"Age vs. Fare by Survival"**.
- Label the x-axis as **'Age'** and the y-axis as **'Fare'**.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
```

AgeFareS...

Submit

```
1 data = pd.read_csv('data/titanic.csv')
2
3
4
5
6
7 # Data Cleaning
8 data['Age'].fillna(data['Age'].median(), inplace=True)
9 data['Embarked'].fillna(data['Embarked'].mode()[0], inplace=True)
10 data.drop('Cabin', axis=1, inplace=True)
11
12 # Convert categorical features to numeric
13 data['Sex'] = data['Sex'].map({'male': 0, 'female': 1})
14 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
15
16 # Write your code here for Scatter Plot for Age vs. Fare by Survived
17
18 colours=data['Survived'].map({0: 'red', 1: 'blue'})
19 plt.scatter(data['Age'], data['Fare'], c=colours)
20 plt.title("Age vs. Fare by Survival")
21 plt.xlabel("Age")
22 plt.ylabel("Fare")
23 plt.show()
```

Average time: 0.369 s 369.00 ms

Maximum time: 0.369 s 369.00 ms

1 out of 1 shown test case(s) passed

Test case 1 369 ms

Debug

Terminal

Test cases

< Prev

Reset

Submit